

The European Space Agency's Juice mission is currently traveling to Jupiter to study its three largest icy moons: Ganymede, Callisto, and Europa. Launched in 2023, the spacecraft will spend years navigating the Jovian system to investigate whether these moons harbor the conditions necessary for life.

Scientists are particularly interested in the vast subsurface oceans believed to exist beneath their thick, frozen crusts. By using advanced spectral sensors and ice-penetrating radar, Juice will map the moons' surfaces and internal structures. This data helps researchers understand the potential habitability of worlds far beyond our solar system's traditional goldilocks zone.